

Knowledge Graph Alignment Under Scarce Supervision: A General Framework With Active Cross-View Contrastive Learning

Weixin Zeng¹, Xiang Zhao¹, Jiuyang Tang, and Changjun Fan

Abstract—Over recent years, a number of knowledge graphs (KGs) have emerged. Nevertheless, a KG can never reach full completeness. A viable approach to increase the coverage of a KG is KG alignment (KGA). The majority of previous efforts merely focus on the matching between entities, while largely neglect relations. Besides, they heavily rely on labeled data, which are difficult to obtain in practice. To address these issues, in this work, we put forward a general framework to simultaneously align entities and relations under *scarce supervision*. Our proposal consists of two main components, relation-enhanced active instance selection (RAS), and cross-view contrastive learning (CCL). RAS aims to select the most valuable instances to be labeled with the guidance of relations, while CCL contrasts cross-view representations to augment scarce supervision signals. Our proposal is agnostic to the underlying entity and relation alignment models, and can be used to improve their performance under limited supervision. We conduct experiments on a wide range of popular KG pairs, and the results demonstrate that our proposed model and its components can consistently boost the alignment performance under scarce supervision.

Index Terms—Active learning (AL), contrastive learning (CL), entity alignment (EA), knowledge graph alignment (KGA).

I. INTRODUCTION

KNOWLEDGE fusion is an important stage of knowledge management, which connects, combines, and updates the knowledge from different knowledge-based systems. With the advent of the era of big data, knowledge graph (KG), as an effective means to extract structured knowledge from massive unstructured data and stocked structured data, becomes an essential part of knowledge management. Taking advantage of data-driven techniques, KGs are usually automatically constructed from individual sources, which might have low coverage [1]. Hence, it calls for establishing the connections among these separately constructed KGs using knowledge fusion techniques, which can in turn lay the foundation for the

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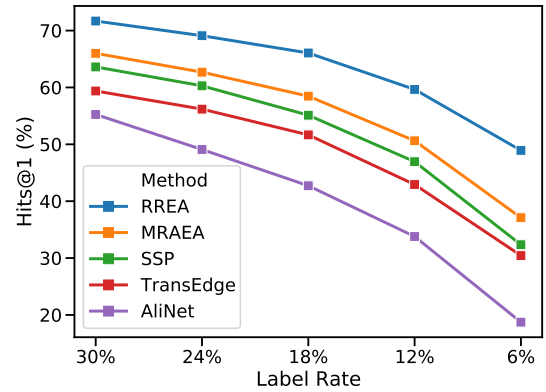


Fig. 1. EA results (i.e., Hits@1) of existing methods [6], [7], [18], [19], [20], [20] on DBP15K_{ZH-EN} [2] when reducing the labeled data. Note that Label Rate refers to the percentage of labeled data. Best viewed in colors.

augmentation and update of KGs. This process is frequently referred to as KG alignment (KGA).

KGA aims to discover the equivalent elements (e.g., entities and relations) in different KGs, thus bridging these individually created KGs. While many works have been dedicated to tackling this problem, the majority merely focus on entity alignment (EA) [2], [3], [4], [5], [6], [7], [8], [9], [10], [11]. They first utilize KG embedding models, such as TransE [12] and GCN (i.e., graph convolutional network) [13], to generate entity embeddings for source and target KGs, respectively [14]. Then, they use the labeled entity pairs to connect KGs and project individual entity embeddings into a unified representation space. Finally, based on the unified entity embeddings, they predict the matched entity pairs. Specifically, for each source entity, they rank the target entities according to the distance between entity embeddings, and the target entity with the smallest distance is predicted to be the match. A few studies investigate the joint alignment of entities and relations [15], [16], [17]. They explicitly conduct the alignment of relations, and use the relation alignment results to improve the alignment of entities. By making better use of relation information, these methods also achieve better overall alignment performance than the counterparts that merely focus on EA. Nevertheless, as the amount of relations is much smaller than that of entities, the joint entity and relation alignment task has yet not aroused great interests.

Noteworthy, the performance of existing KGA solutions is largely dependent on labeled data. As revealed in Fig. 1,

when reducing the amount of labeled data, the performance of the most effective EA approaches all drop significantly. Nevertheless, this issue has been overlooked by current studies. The labeled data that they leverage for training the KGA models only exist in a few commonly used KGs such as DBpedia [21], which, however, are not available in domain-specific KGs that are constructed independently. Consequently, to generate the labeled data, it calls for manual annotation that discovers the equivalent components in different KGs, which is a rather difficult task. Hence, it is crucial to study KGA with scarce supervision and alleviate the dependence on the labeled data.

A few studies have striven to tackle this task [22], [23], [24]. They leverage active learning (AL) to overcome the labeling bottleneck by asking queries in the form of unlabeled instances to be labeled by an oracle (e.g., a human annotator) [25], [26]. Through designing effective query strategies, the active learner can achieve satisfying performance using as few labeled instances as possible, thereby reducing the cost of obtaining labeled data [25]. Nevertheless, these efforts merely focus on entities, neglecting the relations in KGs, and the resultant AL strategies fail to take into account the useful relation information. Additionally, they overlook the abundant unlabeled data that can be used to improve the KG embedding and guide the model training.

Hence, in this work, we propose to address the aforementioned issues. Instead of merely considering EA, we propose to jointly align entities and relations in a unified framework, where the synergy between entities and relations is injected into the KGA process. Concretely, we first put forward a relation-enhanced active instance selection (RAS) strategy, i.e., RAS. Three query strategies are first devised to characterize the informativeness of entities, which take into account the relation information. Then, two approaches are designed to aggregate the query strategies and select the entities to be labeled. One is a reinforced AL framework, which blends query strategies adaptively. The other is a simple weighted combination method for fusing query strategies, which can significantly lower time consumption at the cost of small performance drop compared with the reinforced AL one.

Furthermore, inspired by the recent progress in contrastive learning (CL) [27], [28], we design an unsupervised contrastive loss to exploit the abundant unlabeled data to augment the supervision signals. Specifically, we use two graph representation learning models to capture different views of the structural information of entities and relations, and design a cross-view CL (CCL) strategy, i.e., CCL, to discriminate the embeddings of the same instance in these two views from the embeddings of other instances. By incorporating the unsupervised contrastive loss into the semi-supervised alignment objective, the supervision signals are amplified, which can further ameliorate the alignment performance.

The aforementioned strategies constitute KGA under scarce supervision KGA-SS, a general framework with active CCL to tackle KGA under scarce supervision. To validate the effectiveness and efficiency of KGA-SS, we perform thorough experiments on popular KGA benchmarks against existing baselines. The empirical results prove that KGA-SS manifests

better performance given limited labeled data. Further experiments and analyses also reveal that KGA-SS is a general framework to cope with scarce supervision.

A. Contribution

This article is an extended version of our preliminary work [24]. We make substantial improvements in this extension as follows.

- 1) We extend the original research task (i.e., EA under scarce supervision) to a broader and more general problem (i.e., KGA under scarce supervision) in this version.
- 2) We put forward a general framework with active CCL to address KGA with limited labeled data, where the role of relations and the synergy between entities and relations are exploited to facilitate the alignment process.
- 3) We devise a RAS approach to choose the most valuable instances to be labeled. In addition to the original reinforced AL framework, we put forward a simple weighted combination method to blend the query strategies, which can dramatically reduce the time consumption.
- 4) We design a cross-view CL strategy to distinguish the embeddings in different views, so that the scarce supervision signals can be amplified. The intra-view contrasting is removed to increase the efficiency.
- 5) We conduct a more comprehensive experiment by including some newest benchmarks and making more detailed empirical analysis.

In all, the main contributions of this article are: 1) we put forward KGA-SS, a general framework with active CCL to resolve the scarce supervision issue; 2) we design a RAS strategy to choose valuable instances for labeling, where a reinforced AL approach and a vanilla weighted combination approach are put forward to blend query heuristics, which can facilitate the subsequent alignment; and 3) we are among the first attempts to exploit CL for KGA, which mines supervision signals from the rich unlabeled data to aid the alignment.

B. Organization

In Section II, we introduce related works. In Section III, we describe the preliminaries, including the problem formulation, model overview, and the basic joint entity and relation alignment framework. In Section IV, we introduce the relation-aware AL framework. In Section V, we elaborate the CCL strategy. In Section VI, we conduct thorough experiments and analyze the results, and in Section VII, we conclude this work and discuss future research directions.

II. RELATED WORK

In this section, first we introduce the works related to KGA. Then, we detail solutions designed for KGA under scarce supervision. Finally, we briefly introduce the works related to CL and AL.

A. KG Alignment

Recently, KGA has been intensively studied [29], [30], [31]. Most of the existing KGA literature [2], [32], [33], [34], [35]

are devoted to EA, which detects equivalent entities based on the entity representations. Specifically, some works design graph representation networks to capture the neighboring structural information [20], [36], [37] and produce more expressive representations of entities, and some make use of the side information in KGs, e.g., entity description and attributes, to facilitate EA [19], [38]. Nevertheless, most of these studies merely use relation information to better aggregate the neighboring entity representations [39], [40]. A few works explicitly consider the alignment of relations and investigate the joint entity and relation alignment task [15], [16], [17]. There are some very recent works that aim to address new challenges of KGA (EA), e.g., scalability [41], [42], [43], negative sampling [44], multi-modality [45], etc.

Network alignment has been intensively studied in previous literature [46], which can be considered similar to KGA, as they both aim to address the matching of two graphs. Recent years have also witnessed a rapid progress of network alignment solutions. For instance, Zhao et al. [47] leverage hypergraph to align networks, while Xu et al. [48] tackle the social graph alignment task via sinkhorn distance calculation. However, the solutions to network alignment cannot be directly adapted to tackling KGA, as the graph structures and associated semantics are not the same [49].

B. KGA With Limited Supervision

Labeled data, i.e., aligned entity pairs, are critical to the performance of existing KGA approaches. However, they are difficult to attain in practice. Hence, some works investigate unsupervised EA [50], [51], [52]. They mainly use the side information in KGs, e.g., entity attributes or descriptions, to generate the pseudo labeled data. However, their effectiveness is dependent on the quality of KG side information, which can be unavailable or unevenly distributed in real-life settings [29]. There are some very recent works that investigate the self-supervised setting by optimizing the training process or the alignment inference stage [53], [54], [55]. Notably, a concurrent work [56] puts forward an interactive CL strategy. Nevertheless, it requires the semantic information in KGs, which is not always available in KGA settings.

Additionally, there are several studies that explore the alignment under scarce supervision settings, where a few labeled data are assumed to exist [22], [23], [24]. Berrendorf et al. [22] evaluate the performance of different heuristics of AL in terms of tackling EA. Liu et al. [23] put forward a structure-aware uncertainty sampling strategy to select the informative entities. Zeng et al. [24] design a reinforced AL framework to aggregate query strategies. Besides, they contrast intra-view and cross-view entity embeddings to provide more alignment signals. Nevertheless, these efforts merely focus on entities, neglecting the relations in KGs, and the resultant AL strategies fail to take into account the useful relation information. Additionally, most of them overlook the abundant unlabeled data that can be used to improve the KG embedding and guide the model training. We also notice a research work [57] that studies AL for network alignment. They assume that there exist a set of candidate matching nodes for each node in

question, and develop a relative querying scheme to identify the best questions to ask experts. Such candidate nodes, however, are not available in KGA.

C. AL and CL

AL is a subset of machine learning where a learning algorithm interactively selects the samples to be labeled from unlabeled data so that the desired outputs can be satisfied. It has been widely used in the scarce supervision settings of many tasks such as named entity recognition [58], graph node classification [26], [59], and instance classification [60]. In this work, we exploit the AL framework to tackle KGA with scarce supervision.

CL generates data representations by learning to encode the dissimilarities and similarities among unlabeled samples. It assumes that the vast unlabeled data can also be used as supervision signals to help train the model [61]. CL arrests increasingly more attentions in the field of graph learning [28], [62], [63]. They conduct stochastic augmentation on the input graphs, resulting in two graph views, and then maximize the agreement of the representations in these two views, by which more expressive graph representations can be learned. Hence, we also leverage CL to benefit the KGA process.

III. PRELIMINARIES

In this section, we formulate the research problem. Next, we introduce the model outline. Then, we detail the joint entity and relation alignment framework, which KGA-SS is built upon.

A. Problem Formulation

1) *KG Alignment*: A KG is denoted as $\mathcal{G} = \{(s, r, o)\} \subset \mathcal{E} \times \mathcal{R} \times \mathcal{E}$, where $\mathcal{E}$ is the entity set, $\mathcal{R}$ is the relation set, and a triple (s, r, o) represents a subject entity $s \in \mathcal{E}$ and an object entity $o \in \mathcal{E}$ that are connected by a relation $r \in \mathcal{R}$. The inputs to KGA consist of the source KG $\mathcal{G}_s$, the target KG $\mathcal{G}_t$, a set of labeled entity pairs $\mathcal{S} = \{(u^*, v^*) | u^* \in \mathcal{E}_s, v^* \in \mathcal{E}_t, u^* \Leftrightarrow v^*\}$. $\Leftrightarrow$ denotes equivalence. $\mathcal{E}_s$ and $\mathcal{E}_t$ are the source entity set and target entity set, respectively. Optionally, there could be a set of labeled relation pairs $\mathcal{S}_r = \{(u_r^*, v_r^*) | u_r^* \in \mathcal{R}_s, v_r^* \in \mathcal{R}_t, u_r^* \Leftrightarrow v_r^*\}$, where $\mathcal{R}_s$ and $\mathcal{R}_t$ denote the source relation set and target relation set, respectively. The goal is to discover the equivalent entity and relation pairs in the remaining entities and relations.

2) *KGA With Scarce Supervision*: In this work, we aim to investigate KGA with *scarce supervision*. We assume that there can be a small set of labeled entity pairs with the size of B (also frequently referred to as labeling budget). We also assume that the pre-aligned relation pairs are not provided, since they are usually in much smaller scale compared with entities. In this context, KGA with scarce supervision further includes two sub-tasks, i.e., the selection of entities to label, and KGA with limited labeled data. The former sub-task is defined as selecting B entities from the set of unlabeled entities $\mathcal{U}$ for an oracle to label, so that the labeled samples can be more beneficial to aligning KGs, while the latter is defined in the same way as the traditional KGA problem.

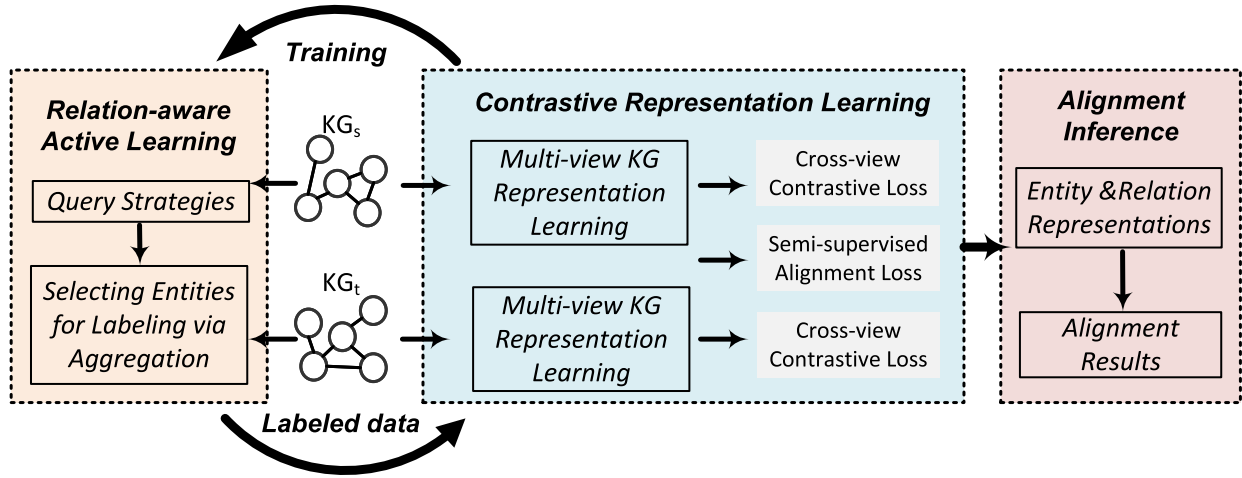


Fig. 2. Framework of our proposal KGA-SS.

B. Model Overview

Fig. 2 describes the overview of KGA-SS. It consists of multiple rounds. In each round, relation-aware AL is first conducted, which utilizes query strategies to characterize the informativeness of instances. Next, either weighted combination or multi-armed bandit (MAB) mechanism is harnessed to fuse the strategies and select the entities to be labeled. The labeled data are then sent to the contrastive representation learning module, where they are used to produce the semi-supervised alignment loss. Notably, the loss also requires (multi-view) KG embeddings as input, which are learned by different KG encoders. The core of this module is the cross-view contrastive loss, which discriminates the structural representations learned in different views and augments the supervision signals from the vast unlabeled data. Eventually, the learned entity and relation representations are harnessed to infer the alignment results.

C. Joint Entity and Relation Alignment

We introduce the joint entity and relation alignment framework in this section, which includes *multi-view representation learning*, *alignment training*, and *relation-enhanced alignment training*. Notably, in this work, we do not aim to put forward a new joint alignment framework. In fact, our proposed solution to weakly supervised KGA is agnostic to entity and relation alignment methods. In other words, KGA-SS is a “plug-in” component to the common KGA frameworks, and it deals specifically with the scarce supervision issue. Hence, we use some baseline alignment models due to their simplicity, which can be replaced with other more advanced methods.

1) *Multi-View Representation Learning*: Since the symbolic features between two KGs cannot be compared directly, we aim to generate entity structural embeddings for alignment. Specifically, we use two basic representation learning models, i.e., GCN [13] and approximate personalized propagation of neural predictions (APP) [64], to model close and distant KG structural information.

Many existing studies on EA have utilized GCN to learn entity embeddings [33], [65]. Given the feature matrix X of

nodes and the KG adjacency matrix A , a two-layer GCN is defined as

$$\mathbf{Z}^{\psi_1} = \text{ReLU}(\hat{A} \text{ReLU}(\hat{A} X W_0) W_1) \quad (1)$$

where $\hat{A}$ is the symmetrically normalized adjacency matrix with self-loops, and $\mathbf{Z}^{\psi_1}$ is the output embedding matrix. W_0 and W_1 are learnable weight matrices.

Despite of the popularity, GCN suffers from the over-smoothing issue when increasing the layers [6]. Thus, we use approximate personalized propagation [64] to model the distant structural information. It is formulated as

$$\mathbf{Z}^{(i)} = (1 - \alpha) \hat{A} \mathbf{Z}^{(i-1)} + \alpha X, \quad i = 1, 2, \dots, k \quad (2)$$

where k is the round of propagation, and α denotes the teleport probability. $\mathbf{Z}^{(0)} = X$, which refers to the initial feature matrix. The output entity embedding matrix is represented as $\mathbf{Z}^{\psi_2}$.

Next, we combine $\mathbf{Z}_\omega^{\psi_1}$ and $\mathbf{Z}_\omega^{\psi_2}$ to generate the final entity representation, i.e., $\mathbf{E}_\omega = \text{agg}(\mathbf{Z}_\omega^{\psi_1}, \mathbf{Z}_\omega^{\psi_2})$, where agg is the aggregation function. $\omega \in \{s, t\}$, where s refers to the source KG and target KG is represented as t .

2) *Alignment Training*: Given the entity embeddings $\mathbf{E}_s$ and $\mathbf{E}_t$, we use a semi-supervised loss to train the representation learning models

$$\mathcal{L}_s = \sum_{(u,v) \in \mathcal{S}} \sum_{(u',v') \in \mathcal{S}'_{(u,v)}} [\text{dis}(u, v) + \gamma - \text{dis}(u', v')]_+ \quad (3)$$

where (u, v) denotes an equivalent entity pair, $\mathcal{S}'_{(u,v)}$ refers to the set of negative samples obtained by corrupting (u, v) using nearest neighbor sampling [66], and $[\cdot]_+ = \max\{0, \cdot\}$. u and v represent entity embeddings, and $\text{dis}(\cdot, \cdot)$ denotes the distance function. γ is the hyperparameter that separates positive samples from negative ones. Such a loss function aims to pull the entity embeddings in the labeled data near, and meanwhile widen the distance between the negative ones.

3) *Relation-Enhanced Alignment*: Since relations can help differentiate entities, we propose to make explicit use of relations to facilitate EA. Specifically, following [15], we approximate relation representations based on the head and tail entity embeddings. Given a relation r , we first retrieve the

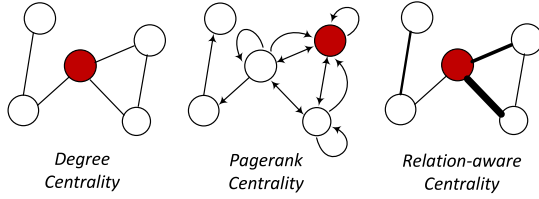


Fig. 3. Illustration of query strategies.

triples that involve it, i.e., $\mathcal{T}_r$. Then, we calculate the average embeddings of all the head and tail entities in $\mathcal{T}_r$, resulting in $\mathbf{H}_r$ and $\mathbf{T}_r$, respectively, which are concatenated and sent to a linear transformation layer to produce the relation embedding.

$$\mathbf{r} = [\mathbf{H}_r; \mathbf{T}_r]\mathbf{W}_r \quad (4)$$

where $[\cdot]$ is the concatenation of two matrices, and $\mathbf{W}_r$ is a learnable weight matrix.

We further integrate the relation embeddings into entity representations to generate the joint entity embeddings. For each entity, we aggregate its adjacent relation representations into the relation context vector $\mathbf{R}_e$, and then concatenate it with the original entity embedding to form the joint entity representation

$$\mathbf{e}_j = [\mathbf{e}; \mathbf{R}_e] \quad (5)$$

where $\mathbf{e}$ denotes the embedding generated by the encoders, and $\mathbf{e}_j$ is the joint entity representation. After obtaining the joint entity representation, we still use (3) for training.

IV. RELATION-AWARE AL

To tackle KGA with limited labeled data, we first investigate how to select more informative entities for annotation (the labeling for relations is not provided as described in the task formulation). Specifically, we adopt the idea of AL to select the entities. The AL process includes multiple iterations. In each iteration, it selects b entities with the highest *informativeness* to label, which are then added to the training set. The iteration ends when the labeling budget B is exhausted.

Next, we put forward the relation-aware AL approach, RAS.

A. Query Strategies

First, we devise several query strategies, i.e., degree centrality, PageRank centrality, and relation-aware centrality to characterize the *informativeness* (more specifically, the representativeness) of entities, which are shown in Fig. 3.

1) *Degree Centrality*: We consider the entities that possess more neighboring entities are of higher value, as knowing the equivalence of these entities can facilitate the alignment of the remaining ones. Hence, we use the degree of an entity node, i.e., degree centrality $\phi_{\text{deg}}(e)$, to characterize the informativeness of an entity.

2) *PageRank Centrality*: We also use the PageRank centrality [67] $\phi_{\text{pr}}(e_i; \Theta_p)$ to characterize the representativeness of entities. Formally

$$\phi_{\text{pr}}(e_i; \Theta_p) = \rho \sum_j A_{ij} \frac{\phi_{\text{pr}}(e_j; \Theta_p)}{\sum_k A_{jk}} + \frac{1 - \rho}{n} \quad (6)$$

where n denotes the number of entities and $\mathbf{A}$ represents the adjacency matrix. ρ denotes the damping parameter and Θ_p refers to the set of parameters.

3) *Relation-Aware Centrality*: To make better use of relations, we devise relation-aware centrality, which attaches greater importance to the entities that are connected with more valuable relations. Specifically, the value of a relation is defined as the reciprocal of its frequency. This is based on the observation that more frequently appearing relations are less differentiable in terms of comparing entities, since these relations might appear in many entities' neighboring structures. In contrast, deriving the equivalence of less frequent relations is of greater help to discriminating entities. Thus, we aim to select the entities that have high degrees (i.e., more connected relations) and meanwhile relations with higher values. The *relation-aware centrality* of an entity is defined as

$$\phi_r(e; \Theta_r) = \sum_{r \in \mathcal{R}_e} \frac{1}{\text{freq}(r)} \quad (7)$$

where $\text{freq}(r)$ is the frequency of relation r , and $\mathcal{R}_e$ is the set of relations associated with entity e .

To make the score values of query strategies comparable, we convert the values to percentiles [68], and $\mathcal{P}_{\text{deg}}$, $\mathcal{P}_{\text{pr}}$, $\mathcal{P}_r$ represent the percentile scores for degree centrality, PageRank centrality, and relation-aware centrality, respectively.

B. Active Entity Selection

In this section, we aggregate the aforementioned query strategies to select the most informative entities for labeling. Specifically, we devise two approaches to make better use of the query strategies, i.e., weighted combination and reinforced active entity selection.

1) *Weighted Combination*: The simplest approach to fuse query strategies and produce a feasible selection method is *weighted combination*, i.e., assigning weights to different strategies and then combine them. Intuitively, query strategies that emphasize entity centrality, i.e., ϕ_{deg} and ϕ_{pr} , should receive higher weights, since the amount of entities is much larger than that of relations. In the experiment, we tune the weights on the validation set.

2) *Reinforced Active Entity Selection*: The significance of the query strategies can vary during the learning process. Hence, in the preliminary version of this work [24], we propose to use MAB to adaptively aggregate the query strategies. We refer interested readers to [24] for more details.

V. CROSS-VIEW CL

Given limited labeled data, we elaborate the cross-view contrastive entity representation learning, CCL, in this section. It aims to mine supervision signals from the vast unlabeled data and further alleviate the scarce supervision issue.

A. Cross-View Contrastive Loss

Inspired by the successful application of CL on unsupervised graph representation learning [27], [69], we put forward a contrastive objective to discriminate the embeddings of the

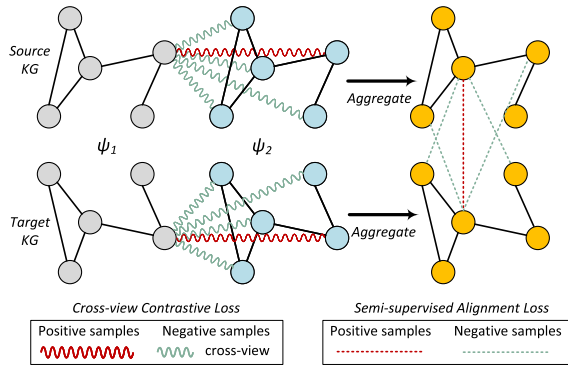


Fig. 4. Illustration of cross-view contrastive loss and semi-supervised loss. Best viewed in colors.

same entity under the two views from the embeddings of other entities, so as to generate more expressive representations by mining supervision signals from the unlabeled data.

Given an entity x_i , we denote the embedding generated by the first and second view as $\psi_1(i)$ and $\psi_2(i)$, respectively. $[\psi_1(i), \psi_2(i)]$ is regarded as a positive sample, and the pairs of embeddings that contain $\psi_1(i)$ [or $\psi_2(i)$] and the embedding of another entity in another view are regarded as the negative ones. Then, we adopt the InfoNCE loss [70] to calculate the cross-view contrastive loss of the entity in the view ψ_1 , which is denoted as $\ell^{\psi_1}(x_i)$. Fig. 4 provides detailed illustrations. The contrastive object of the second view $\ell^{\psi_2}(x_i)$ is defined similarly. Hence, the overall cross-view contrastive loss is defined as $\mathcal{L}_u = (1/2(n_s + n_t)) \sum_{i=1}^{n_s+n_t} [\ell^{\psi_1}(e_i) + \ell^{\psi_2}(e_i)]$, where the amount of entities in the two KGs are denoted as n_s and n_t , respectively.

B. Optimized Training

We aggregate the semi-supervised alignment loss in (3) and the cross-view contrastive loss, leading to the final loss function for training the KGA model

$$\mathcal{L} = \mathcal{L}_s + \lambda_u \mathcal{L}_u \quad (8)$$

where $\lambda_u > 0$ denotes the loss-balancing hyperparameter, which is tuned on the validation set.

C. Alignment Inference

Based on the learned unified embeddings, we infer the equivalence of entities and relations.

For each source entity, we calculate the distances with all target entities according to a given distance metric, e.g., Euclidean distance. The closest target entity is regarded as the match.

As for the inference of relation alignment, in addition to the distance between relation embeddings, we follow [15] and use the statistical information known in advance to improve the distance calculation. We assume that the more aligned entities exist in the entities that are connected to the two relations, the more likely these two relations are equivalent. Hence, for a pair of relations r_1 and r_2 , we collect the pre-aligned entities existing in the head or tail entities of these two

relations and denote as $\mathcal{P}_{r_1 r_2} = \{(e_{i1}, e_{i2}) | e_{i1} \in \mathcal{HT}_{r_1}, e_{i2} \in \mathcal{HT}_{r_2}, (e_{i1}, e_{i2}) \in \mathcal{S}\}$. $\mathcal{HT}_{r_1}$ and $\mathcal{HT}_{r_2}$ denote the sets of head or tail entities for relation r_1 and r_2 , respectively. Thus, the adapted distance between two relations is calculated as

$$\text{dis}(r_1, r_2) = \|\mathbf{r}_1 - \mathbf{r}_2\|_{L_1} - \beta \frac{|\mathcal{P}_{r_1 r_2}|}{|\mathcal{HT}_{r_1} \cup \mathcal{HT}_{r_2}|} \quad (9)$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are relation embeddings, β is the hyper-parameter determining the influence of statistical information.

The overall algorithm of KGA-SS is described in Algorithm 1.

Algorithm 1 Active KGA

Input : $\mathcal{G}_s$: source KG; $\mathcal{G}_t$: target KG; $\mathcal{S}$: labeled data; B : budget; b : the number of entities selected in each round; An oracle that labels data.

Output: $\mathcal{M}$: aligned entities and relations.

- 1 $\mathcal{S}' \leftarrow \mathcal{S}$;
 - 2 **while** $|\mathcal{S}'| - |\mathcal{S}| < B$ **do**
 - 3 Use AL strategies to select b entities for labeling, resulting in $\mathcal{S}_b$ (see Section IV);
 - 4 $\mathcal{S}' \leftarrow \mathcal{S}' \cup \mathcal{S}_b$;
 - 5 Generate entity and relation representations using graph encoders (see Section III-C);
 - 6 Calculate $\mathcal{L}_s$ using $\mathcal{S}'$ via Equation (3);
 - 7 Calculate $\mathcal{L}_u$ and obtain the overall loss $\mathcal{L}$ for training unified representations (see Section V);
 - 8 Infer $\mathcal{M}$ using the learned representations (see Section V-C);
 - 9 **return** $\mathcal{M}$;
-

VI. EXPERIMENT

In the experiment, we conduct extensive evaluations and aim to answer the following research questions.

- 1) *RQ1*: Does KGA-SS attain better performance than the baseline models in terms of KGA with scarce supervision?
- 2) *RQ2*: Are the proposed modules, i.e., RAS, CCL, and relation modeling, beneficial to the overall alignment results?
- 3) *RQ3*: Is KGA-SS model-agnostic? Can it be used on top of other methods to enhance their capability of tackling scarce supervision?
- 4) *RQ4*: Why is the CL component effective? Is combining the embeddings from different views useful enough?
- 5) *RQ5*: Can the proposed AL method be used on other models? Does it achieve better performance than using the query strategies separately? Weighted combination and reinforced AL, which one is better?

A. Experimental Settings

1) *Datasets*: We use three popular KGA datasets for evaluation.

TABLE I
STATISTICS OF THE DATASETS

Dataset	KG pairs	#Ents	#Rels	#Triples
DBP15K _{ZH-EN}	DBpedia (Chinese&English)	38,960	3,024	165,556
DBP15K _{JA-EN}	DBpedia (Japanese&English)	39,594	2,452	170,698
DBP15K _{FR-EN}	DBpedia (French&English)	39,654	2,111	221,720
SRPRS _{EN-FR}	DBpedia (English&French)	30,000	398	70,040
SRPRS _{EN-DE}	DBpedia (English&German)	30,000	342	75,740
SRPRS _{DBP-WD}	DBpedia & Wikidata	30,000	397	78,580
SRPRS _{DBP-YG}	DBpedia & YAGO	30,000	253	70,317
OpenEA _{E-F-V1}	DBpedia (English&French)	30,000	477	88,198
OpenEA _{E-F-V2}	DBpedia (English&French)	30,000	359	176,430
OpenEA _{E-D-V1}	DBpedia (English&German)	30,000	346	98,095
OpenEA _{E-D-V2}	DBpedia (English&German)	30,000	265	177,499

- 1) DBP15K [2],¹ including three cross-lingual KG pairs extracted from DBpedia.
- 2) SRPRS [36],² comprising two cross-lingual and two mono-lingual KG pairs extracted from DBpedia, Wikidata, and YAGO.
- 3) OpenEA [49],³ which consists of cross-lingual KG pairs (i.e., English to French DBpedia, and English to German DBpedia) with different density. The denser KG pairs are denoted with suffix “-V2,” while the relatively sparser ones are denoted with suffix “-V1.”

Table I provides more details of the datasets. For all datasets, the testing:validation:training ratio is 7:1:2. As we investigate KGA with scarce supervision, the initial training set only consists of 500 labeled entity pairs. Next, in each round of the AL process, entities from the remaining training data are selected for annotation and added to the initial training set.

2) *Implementation Details:* Concerning the GCN graph encoder, we adopt two two-layer GCNs [33], [65]. As to APP, we follow the original settings and set α to 0.2, the round of propagation k to 5 [64]. The dimensionality is set to 100. In alignment training [see (3)], $\text{dis}(\cdot, \cdot)$ is set to Euclidean distance, and γ is selected among [1, 3, 5, 10]. Embedding aggregation is implemented as $\text{agg}(\mathbf{Z}_\omega^{\psi_1}, \mathbf{Z}_\omega^{\psi_2}) = \lambda_e \mathbf{Z}_\omega^{\psi_1} + (1 - \lambda_e) \mathbf{Z}_\omega^{\psi_2}$, where $\lambda_e \in (0, 1)$ is selected among [0.2, 0.4, 0.6, 0.8]. β in the relation modeling is set to 20 [15].

As to AL, we set the labeling budget b for each round to 50. The damping parameter in (6) is set to 0.85. We select λ_u in (8) among [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35] and use the Adam optimizer. In the alignment inference process, we also use the Euclidean distance.

After tuning on the validation set, we set γ to 1, λ_e to 0.2, λ_u to 0.2, the weights of degree centrality, PageRank centrality, and relation-aware centrality to 0.4, 0.4, 0.2. Noteworthy, unless specified, we use the newly proposed weighted combination strategy for fusing query strategies in KGA-SS. We will compare weighted combination and reinforced AL strategies in Section VI-F.

We conducted the experiments on a personal computer with Ubuntu system, an NVIDIA GeForce GTX TITAN X GPU, Intel Core i7-4790 CPU with 32 GB memory. The experiments

TABLE II
HITS@1 VALUES OF EA ON DBP15K AND SRPRS
UNDER DIFFERENT BUDGETS

Budget	Method	DBP15K			SRPRS			
		ZH-EN	JA-EN	FR-EN	EN-FR	EN-DE	DBP-WD	DBP-YG
B=500	GCN	22.5±0.4	20.9±0.4	23.0±0.5	14.7±0.4	30.6±0.3	15.6±0.2	21.6±0.2
	APP	24.7±0.3	24.3±0.5	25.8±0.7	17.0±0.3	31.6±0.3	17.4±0.2	23.4±0.4
	ALEA	27.4±0.1	25.9±0.1	26.9±0.1	18.9±0.2	32.5±0.1	19.7±0.1	25.3±0.2
	RAC	28.6±0.3	28.7±0.6	29.4±0.2	19.9±0.2	34.1±0.3	20.7±0.3	27.3±0.2
	GCN-rel	27.9±0.5	26.8±0.5	28.1±0.8	17.4±0.2	34.0±0.4	17.6±0.5	23.6±0.6
	APP-rel	27.7±1.0	28.3±0.6	28.9±0.8	18.4±0.2	33.6±0.4	18.2±0.1	25.9±0.2
B=1500	KGA-SS	31.8±0.7	29.9±0.5	31.1±0.4	24.0±0.3	38.6±0.1	23.0±0.3	30.0±0.3
	GCN	31.5±0.4	32.1±0.6	32.5±0.4	20.5±0.3	35.6±0.2	22.4±0.4	27.6±0.3
	APP	33.1±0.4	35.3±0.4	34.8±0.4	22.0±0.3	36.5±0.1	23.5±0.2	28.8±0.3
	ALEA	35.4±0.2	36.8±0.1	36.0±0.1	23.7±0.1	37.2±0.1	24.8±0.1	30.5±0.1
	RAC	37.7±0.1	39.5±0.2	38.5±0.3	25.5±0.2	39.3±0.4	26.7±0.2	32.7±0.1
	GCN-rel	38.5±0.9	37.9±0.5	38.7±0.4	22.7±0.4	40.1±0.4	23.8±0.4	29.1±0.3
	APP-rel	37.5±0.4	38.9±0.4	38.3±0.2	22.9±0.4	37.7±0.7	23.5±0.1	30.6±0.3
	KGA-SS	41.2±0.7	42.1±0.1	42.1±0.2	27.1±0.3	43.6±0.3	27.3±0.3	33.6±0.2

were conducted for five independent runs, and the average results, along with the standard deviations, are reported in the following.

3) *Evaluation Metrics:* Following previous works [32], for each test entity (resp., relation) in the source KG, the target entities (resp., relations) are ranked ascendingly based on the embedding distance as in Section III-C, and the corresponding Hits@1 value is used as the evaluation metric. It is defined as the percentage of source entities (resp., relations) whose ground-truth target entity (resp., relation) is ranked first. Unless specified, the Hits@1 values are denoted as percentages.

4) *Compared Methods:* Unlike most of existing EA solutions that design more effective representation learning networks to capture the neighboring structures, KGA-SS is devised specifically to address the scarce supervision issue. It is agnostic to the underlying representation learning models, and can be used upon these models to improve their ability of tackling limited labeled data. Therefore, we mainly compare KGA-SS with solutions designed for addressing scarce supervision. Specifically, we compare KGA-SS with ALEA, which leverages the degree centrality as the query strategy in its active EA framework [22], as well as RAC, our preliminary work that does not explicitly exploit the relation information. We do not include ActiveEA for comparison, since it adopts a different running mechanism and cannot be directly compared [23].

For the comprehensiveness of evaluation, we also compare with two baseline representation learning models (that are used in this work), i.e., GCN and APP. Besides, we also integrate the relation alignment module described in Section III-C into GCN and APP, resulting in GCN-rel and APP-rel, and report the corresponding results in the following.

To validate the generality of KGA-SS, we employ it on CEA [65], a representative EA model that leverages auxiliary information, and also on one of the most effective representation learning models RREA [6], in Section VI-D.

B. Main Results (RQ1)

We first present the EA results. Then we introduce the performance of relation alignment.

1) *EA Results:* First, we report the results of EA in Tables II and III, where the labeling budget B is set to 500 and

¹The dataset can be accessed via <https://github.com/nju-websoft/JAPE>.

²The dataset can be accessed via <https://github.com/nju-websoft/RSN>.

³The dataset can be accessed via <https://github.com/nju-websoft/OpenEA>.

TABLE III
HITS@1 VALUES OF EA ON OPENEA UNDER DIFFERENT BUDGETS

Budget	Method	V1		V2	
		EN-FR	EN-DE	EN-FR	EN-DE
B=500	GCN	20.8±0.1	50.9±0.1	32.7±0.2	65.8±0.2
	APP	22.1±0.3	49.9±0.2	37.8±0.3	65.4±0.5
	ALEA	20.8±0.2	50.9±0.1	32.8±0.1	65.8±0.2
	RAC	25.3±0.4	52.8±0.2	38.8±0.3	69.0±0.2
	GCN-rel	23.7±0.4	51.5±0.3	40.9±0.5	66.9±0.4
	APP-rel	25.1±0.6	51.1±0.5	38.4±0.4	63.6±0.3
B=1500	KGA-SS	29.5±0.2	55.6±0.2	43.8±0.3	70.3±0.3
	GCN	29.2±0.2	55.1±0.1	43.2±0.2	72.6±0.1
	APP	30.8±0.3	55.7±0.3	47.8±0.2	71.4±0.1
	ALEA	29.3±0.3	55.1±0.1	43.2±0.2	72.6±0.1
	RAC	35.4±0.2	58.7±0.1	49.6±0.5	75.0±0.2
	GCN-rel	31.1±0.5	57.9±0.4	47.7±0.7	75.6±0.2
	APP-rel	32.4±0.4	56.4±0.5	47.2±0.4	69.1±0.3
	KGA-SS	36.2±0.2	60.5±0.1	48.0±0.2	76.6±0.1

TABLE IV
HITS@1 RESULTS OF RELATION ALIGNMENT UNDER DIFFERENT BUDGETS. WE ONLY REPORT RESULTS ON TWO KG PAIRS FOR EACH DATASET. THE PERFORMANCE ON THE REMAINING KG PAIRS EXHIBITS SIMILAR TRENDS AND IS OMITTED IN THE INTEREST OF SPACE

Budget	Method	DBP15K		SRPRS		15K(V1)	
		ZH-EN	JA-EN	EN-FR	EN-DE	EN-FR	EN-DE
B=500	GCN-rel	35.7±1.6	32.1±2.3	45.7±2.1	52.9±2.8	53.8±2.6	55.5±0.7
	APP-rel	34.2±0.7	29.2±2.5	48.0±1.1	59.0±2.8	53.9±2.7	53.5±0.7
	KGA-SS	37.5±1.2	33.0±0.9	51.6±1.0	55.9±1.8	55.9±1.4	53.6±2.5
B=1500	GCN-rel	48.4±1.3	41.6±1.3	52.4±2.8	61.4±2.0	63.3±2.5	58.7±3.9
	APP-rel	44.2±0.6	39.4±0.6	53.5±1.8	61.4±1.8	62.7±2.6	60.0±3.4
	KGA-SS	46.4±1.2	42.7±0.5	57.8±1.1	61.8±1.4	64.9±0.9	62.8±0.7

1500, respectively. It can be observed that methods exploiting AL framework attain superior performance than the baseline models (i.e., APP and GCN) on all datasets. Particularly, KGA-SS significantly outperforms the baseline models (over 40% on SRPRS_{EN-FR}), showcasing the effectiveness of our proposal given scarce supervision.

Next, we compare KGA-SS (and RAC) with ALEA. Evidently, our proposal is more performant than current AL-based EA approaches under scarce supervision. The superiority mainly comes from the two components, i.e., RAS and CCL, which will be analyzed later.

Besides, making better use of the relation information, KGA-SS consistently achieves better results than RAC over all datasets, validating the effectiveness of exploiting the synergy between entities and relations. This is also reflected in the fact that the results of GCN-rel (respectively, APP-rel) largely outperform those of GCN (respectively, APP).

2) *Relation Alignment Results*: KGA-SS can inherently tackle the alignment of relations due to the characterization of relation representations. Thereby, we report the relation alignment results in Table IV. It observes that, overall speaking, KGA-SS achieves superior relation alignment performance than the baseline models, mainly due to the better embeddings induced by the more advanced training mechanisms (e.g., CL and AL). Nevertheless, it should be noted that these new

mechanisms are still mainly designed to accurately align entities, and hence the relation alignment results of KGA-SS are not always satisfying. For instance, on denser datasets that contain a large amount of triples, the relation representations would become less accurate, and our entity selection strategy would further amplify this issue, leading to worse alignment performance. Hence, it calls for more delicate modeling of relation information, as well as training strategies that can better incorporate the relations, which we will leave as the future work.

C. Ablation Results (RQ2)

To validate the effectiveness of the proposed components in KGA-SS, i.e., cross-view contrastive loss, the AL strategy, and the modeling of relation information, we perform the following ablation study. Specifically, we select B among [0, 250, 500, 750, 1000, 1250, 1500, 1750, 2000], and obtain the corresponding entity and relation alignment results of KGA-SS, KGA-SS w/o AL, KGA-SS w/o CL, and KGA-SS w/o r(elation modeling), where AL denotes using our proposed relation-aware AL strategy, CL denotes using our proposed CCL strategy, and r(elation modeling) denotes the integration of relation information.

It reads from Fig. 5(a) that, overall speaking, all three strategies are beneficial to EA. More concretely, the modeling and integration of relation information is critical to the alignment results in almost all cases, since it explicitly introduces additional signals for alignment from relations, which can help discover equivalent entities. Besides, it is observed that, the AL strategy is more effective when B is smaller, while CL brings more benefits when B becomes larger. As to the ablation study on relation alignment, it reveals in Fig. 5(b) that adopting our proposed AL strategy RAS consistently leads to better performance on DBP15K and SRPRS_{EN-FR}, while on the remaining datasets, the AL strategy only brings superior results when there is a larger amount of labeled data. This is because, although taking into account the relation information, RAS still prioritizes entities with higher degrees (i.e., those with more connected relations), which, however, provide less useful signals for detecting the equivalence of relations. This issue would further amplify on relatively denser datasets, e.g., OpenEA. That being said, it should be noted that RAS boosts the alignment performance of entities, which are of much higher quantity compared with relations, as shown in Table I. Regarding the CL strategy, it seems to have trivial influence on relation alignment. This can be ascribed to the fact that our proposed CL strategy CCL only contrasts the entity embeddings that are directly generated from the representation learning model, which has little effect on the alignment of relations.

D. Generality of KGA-SS (RQ3)

We further testify the generality of KGA-SS by applying it on top of two representative alignment strategies.

First, we apply KGA-SS on CEA [65], which utilizes entity names to complement KG structure in pursuit of more accurate alignment. As shown in Table V, KGA-SS can enhance the

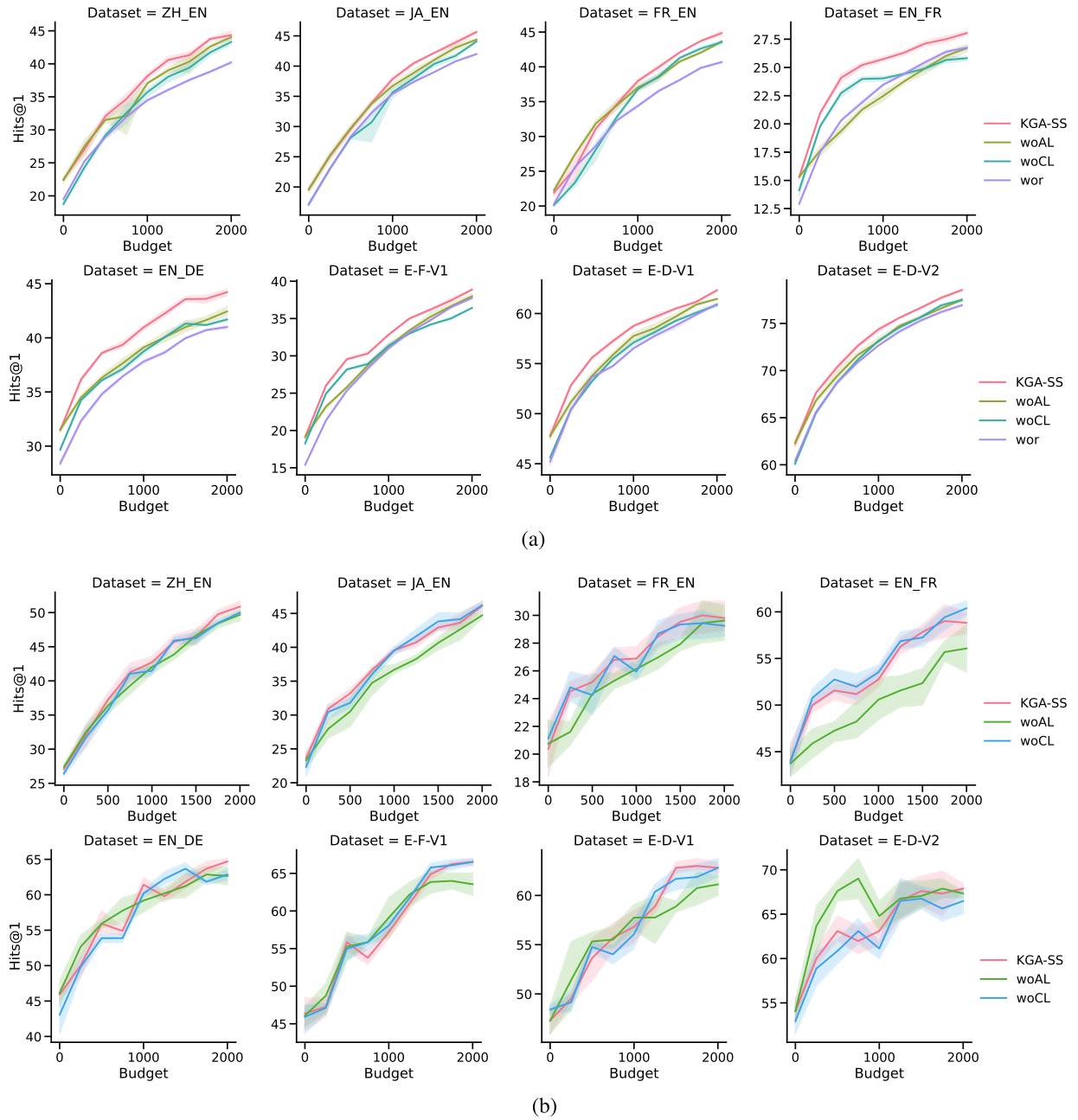


Fig. 5. Hits@1 values of (a) EA results and (b) relation alignment results in ablation study. The shaded area in this figure and the following ones denotes the standard deviation. Best viewed in colors.

TABLE V

HITS@1 EA RESULTS OF APPLYING KGA-SS ON CEA AND RREA

B	Method	DBP15K _{ZH-EN}	SRPRS _{EN-FR}	DBP15K _{JA-EN}	SRPRS _{EN-DE}
500	CEA	64.2±0.1	90.1±0.2	70.6±0.3	87.1±0.4
	+KGA-SS	69.2±0.3	91.0±0.5	74.8±0.3	87.7±0.3
1500	CEA	68.4±0.2	91.9±0.3	75.2±0.2	88.3±0.2
	+KGA-SS	72.4±0.3	92.2±0.3	78.0±0.2	88.7±0.1
500	RREA	48.4±0.4	32.1±0.4	35.1±0.3	17.1±0.3
	+KGA-SS	52.2±0.2	35.9±0.4	38.3±0.3	21.9±0.4
1500	RREA	57.6±0.3	37.3±0.4	41.7±0.1	22.7±0.2
	+KGA-SS	60.7±0.6	39.7±0.2	44.9±0.1	28.3±0.2

alignment results of CEA, and the improvement is more prominent on DBP15K_{ZH-EN} and DBP15K_{JA-EN} where the quality of entity names is closer to real-life situations.

Besides, we apply KGA-SS on RREA [6], one of the most performant EA methods, so as to check whether KGA-SS would improve its performance under scarce supervision. Specifically, we re-implement the model by following the description in the original paper and apply KGA-SS on it. Since RREA already explicitly aggregates the relation information when generating the entity representation, we omit the relation-enhanced alignment in Section III-C to avoid repetition. As shown in Table V, using KGA-SS can also boost the results of RREA, which verifies its generality.

E. Experiments on CL (RQ4)

We aim to testify the usefulness of CCL in this section. First, we evaluate the performance of merely aggregating

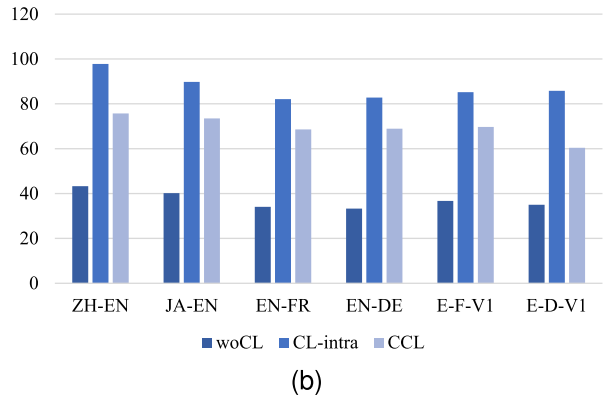
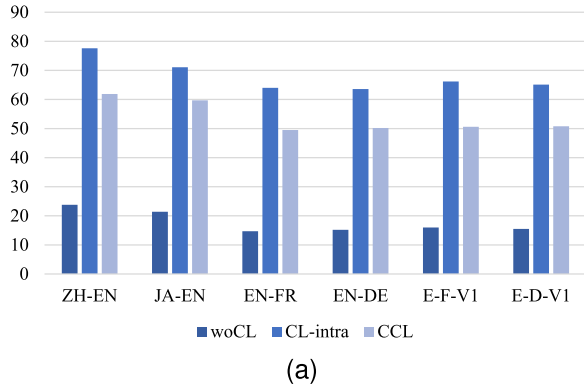


Fig. 6. Running time comparison of CL strategies (in seconds). (a) $B = 500$. (b) $B = 1500$.

TABLE VI
HITS@1 EA VALUES OF VARIANTS OF CL AFTER
REMOVING THE INFLUENCE OF AL

Method	DBP15K _{ZH-EN}		DBP15K _{JA-EN}		SRPRS _{EN-FR}	
	B=500	B=1500	B=500	B=1500	B=500	B=1500
GCN-rel	28.1±0.6	38.3±0.8	25.0±0.3	37.8±0.7	14.6±0.3	20.2±0.3
APP-rel	27.6±0.4	36.9±0.7	27.3±0.8	39.0±0.3	16.7±0.1	22.0±0.3
Comb..	28.5±0.4	38.5±1.2	27.1±1.9	39.5±0.2	18.5±0.3	23.1±0.6
+ CL-intra	30.7±0.3	40.5±0.4	29.7±0.5	40.4±0.6	18.9±0.3	24.5±0.4
+ CCL	31.5±0.3	40.8±1.0	29.7±0.6	41.1±0.4	19.4±0.4	24.9±0.6
$\lambda_u = 0.01$	29.6±0.2	39.4±0.6	29.3±0.5	40.1±0.4	19.1±0.3	24.1±0.5
$\lambda_u = 0.03$	31.4±0.2	40.8±0.2	29.7±0.7	40.9±0.4	19.4±0.5	24.6±0.5
$\lambda_u = 0.10$	31.9±0.3	40.9±0.2	30.4±0.6	40.7±0.4	19.6±0.3	25.2±0.4
$\lambda_u = 0.15$	31.9±0.5	40.7±0.3	30.3±0.8	40.5±0.4	19.5±0.3	25.2±0.2
$\lambda_u = 0.20$	31.6±0.4	40.1±0.3	29.9±0.7	40.1±0.6	19.3±0.4	25.0±0.3

Note that + CL-intra is equivalent to $\lambda_u = 0.05$ in the bottom half of the table.

TABLE VII
HITS@1 EA VALUES OF VARIANTS OF CL AFTER REMOVING
THE INFLUENCE OF AL (MORE RESULTS)

Method	SRPRS _{EN-DE}		OpenEA _{E-F-V1}		OpenEA _{E-D-V1}	
	B=500	B=1500	B=500	B=1500	B=500	B=1500
GCN-rel	30.3±0.3	35.7±0.1	19.9±0.2	29.0±0.3	49.2±0.3	55.1±0.3
APP-rel	31.2±0.2	36.3±0.2	22.2±0.2	30.8±0.3	49.9±0.2	55.7±0.4
Comb..	34.4±0.3	39.1±0.5	25.0±0.4	32.7±0.5	52.2±0.3	58.3±0.4
+ CL-intra	35.4±0.4	40.9±0.4	24.9±0.3	34.4±0.4	53.1±0.5	59.2±0.4
+ CCL	36.3±0.3	41.0±0.5	25.8±0.2	35.2±0.6	53.8±0.4	59.7±0.4
$\lambda_u = 0.01$	35.6±0.4	40.2±0.5	25.8±0.3	33.8±0.4	53.0±0.2	58.9±0.3
$\lambda_u = 0.03$	36.2±0.4	41.1±0.8	25.7±0.4	34.9±0.5	53.6±0.4	59.6±0.4
$\lambda_u = 0.10$	36.0±0.2	40.9±0.3	25.8±0.4	35.0±0.6	53.4±0.2	59.5±0.4
$\lambda_u = 0.15$	35.5±0.2	40.7±0.3	25.7±0.6	34.7±0.5	52.8±0.3	59.2±0.4
$\lambda_u = 0.20$	35.0±0.2	40.5±0.3	25.5±0.6	34.3±0.6	52.2±0.5	58.8±0.3

the embeddings. Then we analyze the efficiency of the CL strategies. Finally, we conduct parameter analysis.

1) *Comparison With Mere Combination of Embeddings:* In Tables VI and VII, we remove AL from KGA-SS, and provide the Hits@1 values of GCN-rel, APP-rel, Comb. (i.e., the combination of these two embeddings), and Comb.+CCL (i.e., adding CCL on Comb.).

From the tables, it observes that, combining the embeddings only brings minor improvement, and even lowers the results in some cases such as $B = 500$ on DBP15K_{JA-EN}. In comparison, when introducing CCL, Comb.+CCL attains much better results on all settings. This validates that the CCL is what really works.

We further compare the CCL strategy CCL with the one proposed in our preliminary work, denoted as CL-intra. The superior performance of CCL demonstrates the effectiveness of merely modeling the cross-view contrastive loss. This is partially due to the fact that the embeddings in a single view are already discriminating, and it is less effective to further exploit the intra-view loss to aid the learning process.

2) *Efficiency Analysis:* We further compare the running time to examine the efficiency of CCL. Fig. 6 unveils that, although beneficial to the overall performance, introducing CL also leads to higher running time cost than the baseline model, mainly due to the additional contrasting process. Plus, compared with our preliminary model CL-intra, CCL reduces the time cost across all datasets, showcasing the efficiency of the CCL.

3) *Sensitivity Analysis:* We conduct sensitivity analysis on the hyperparameter λ_u , which represents the weight of cross-view contrastive loss in the overall loss function. Tables VI and VII show that the alignment results are generally stable when varying λ_u . Besides, on DBP15K and SRPRS, a larger λ_u is preferred, while a relatively smaller λ_u is required on OpenEA.

F. Experiments on AL (RQ5)

We analyze the AL component RAS in this section.

1) *Effectiveness of RAS on Baseline Models:* We apply RAS on the baseline models, i.e., GCN-rel and APP-rel, and report the EA and relation alignment (RA) results in Figs. 7 and 8. We also include a popular AL strategy for comparison, i.e., degree centrality, which is denoted as deg.

It shows in the figures that the EA performance of both GCN-rel and APP-rel are improved after using RAS across all datasets. The improvement is more prominent when the budget is smaller. Besides, the improvement brought by RAS is more notable compared with deg. As to the *relation alignment* results, it can be observed that adopting deg brings worse performance than the baseline models on most datasets. This is because this strategy merely considers the centrality of entities and neglects the role of relations. Reversely, adopting our proposed relation-aware AL strategy RAS can still improve the relation alignment results of baseline models on the majority of datasets, except for the dense dataset such

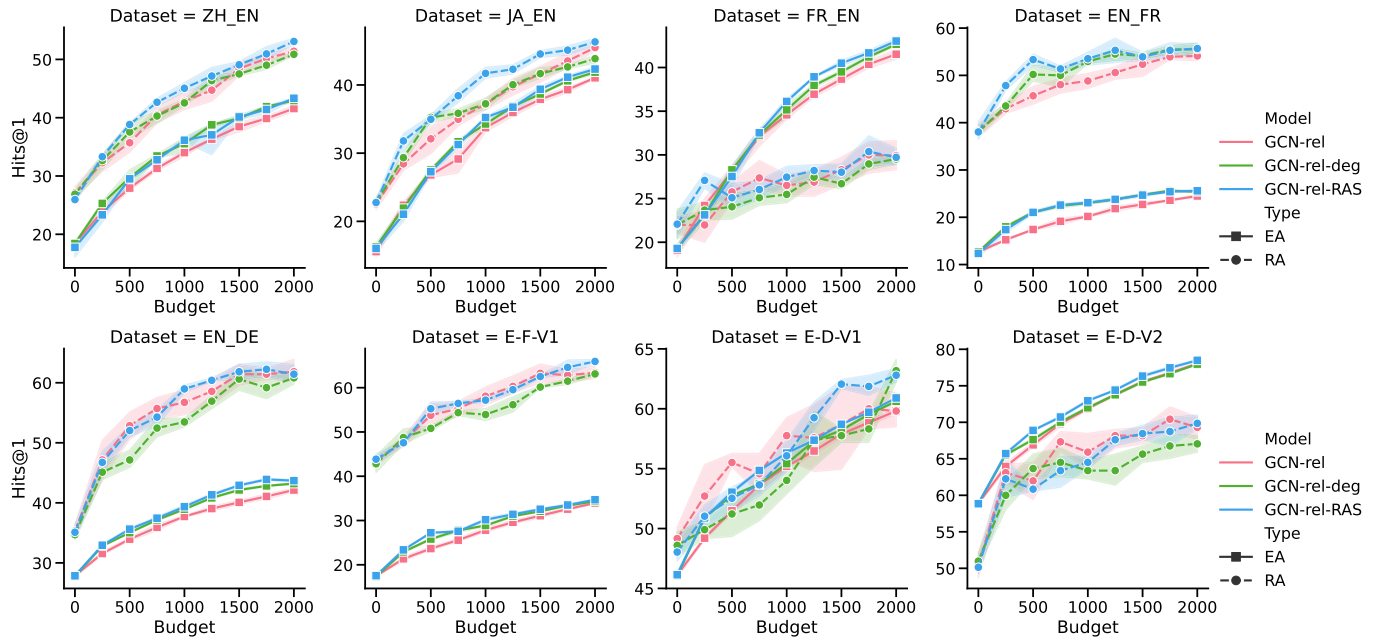


Fig. 7. Hits@1 values of applying AL on baseline models (GCN). Best viewed in colors.

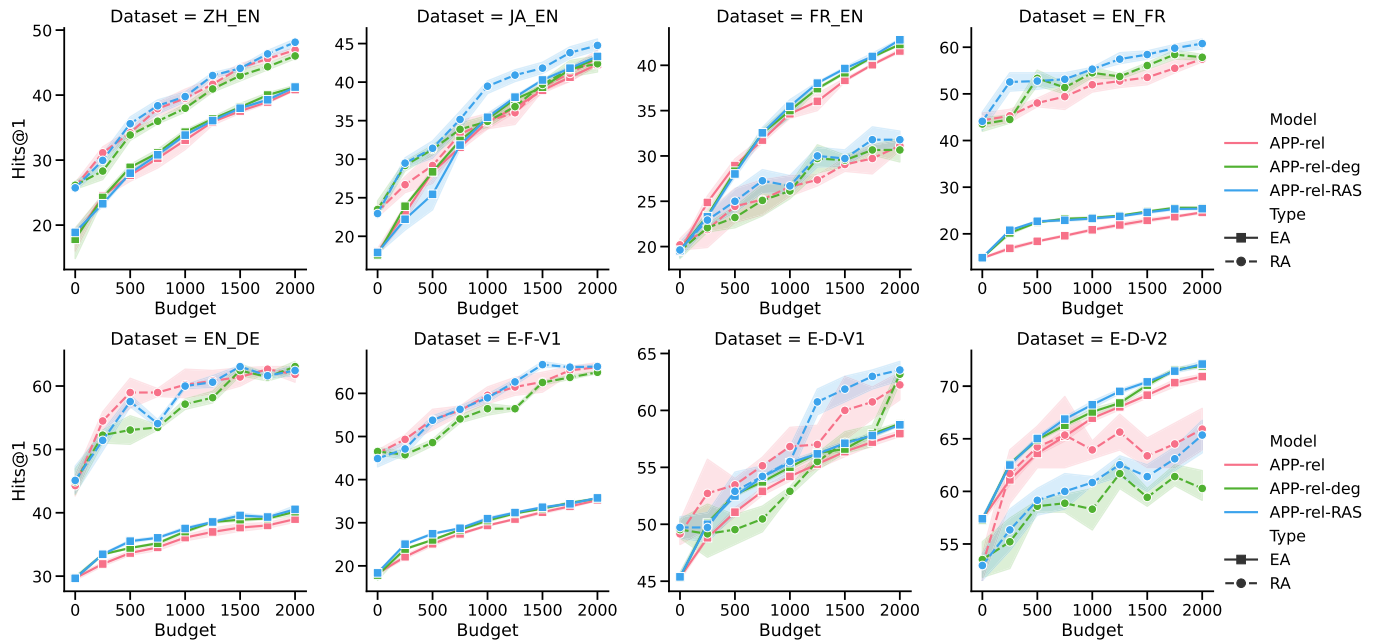


Fig. 8. Hits@1 values of applying AL on baseline models (APP). Best viewed in colors.

as $\text{OpenEA}_{\text{E-D-V2}}$ where the relation representations are less accurate, as each relation is associated with too many triples. Overall, exploiting RAS can improve the KGA performance of baseline models.

2) *Comparison With Using Query Strategies Separately:* We replace RAS in KGA-SS with the degree (deg), PageRank (pr) and relation-aware (rel) centrality, respectively. It observes from Fig. 9 that, overall speaking, KGA-SS achieves the best balance between entity and relation alignment performance. Specifically, KGA-SS attains the best overall EA results, despite that pr outperforms KGA-SS in some cases, e.g.,

$\text{SRPRS}_{\text{EN-DE}}$. Besides, although rel attains the leading performance on relation alignment, its EA results are the worst. In comparison, KGA-SS still attains good results on relation alignment, which are only second to rel.

3) *Comparison With Reinforced AL:* Next, we compare the fusion strategies, i.e., our newly proposed weighted combination strategy (KGA-SS) and the reinforced AL via MAB, denoted as mab. As shown in Fig. 9, the performance of KGA-SS and mab are very close, with mab slightly surpassing KGA-SS. Nevertheless, the time cost of mab (approximately 40 min for each round) is much higher than

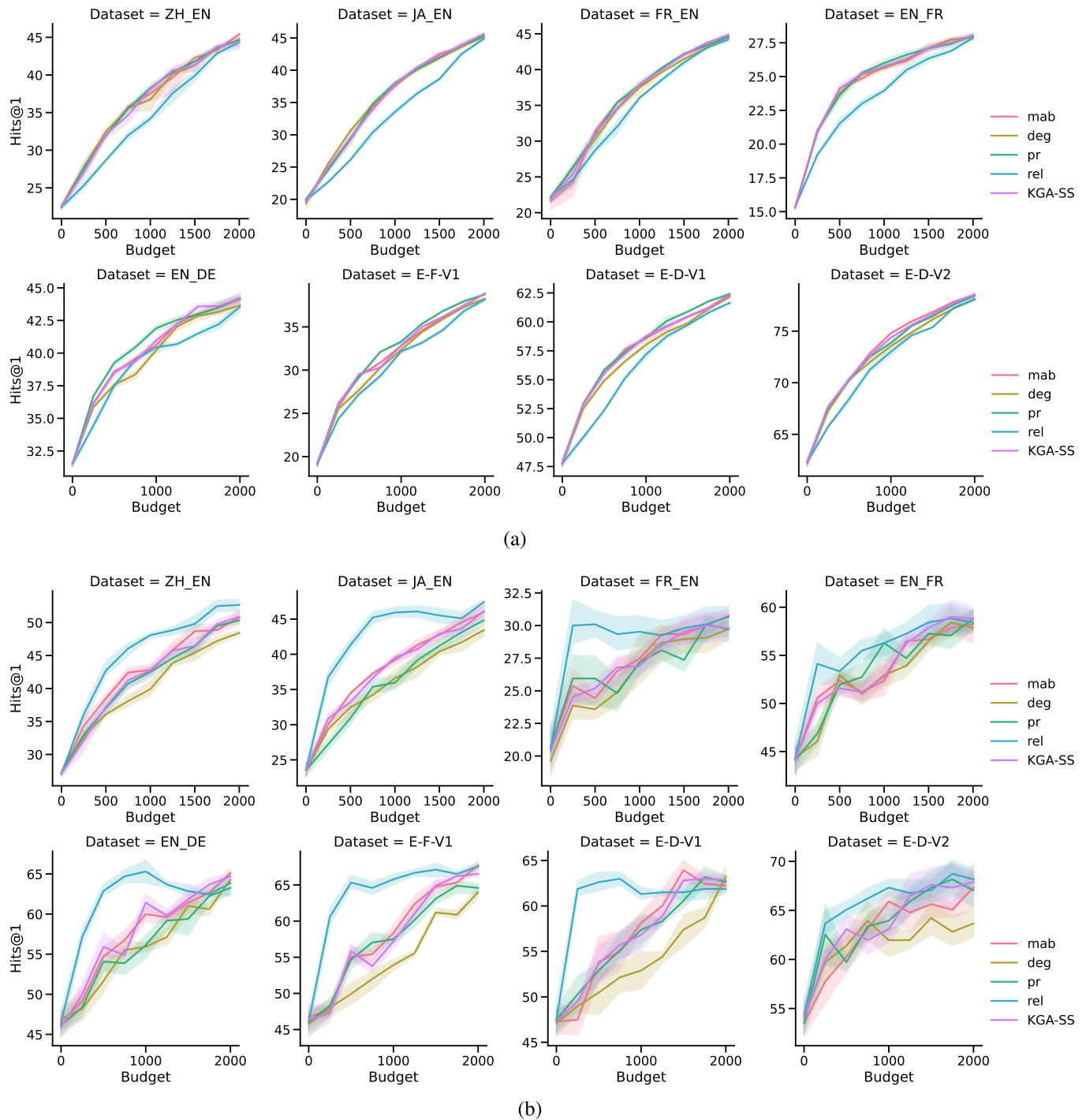


Fig. 9. Hits@1 values of (a) EA results and (b) relation alignment results of the variants of AL strategy. Best viewed in colors.

weighted combination (approximately 80 s for each round) due to the reinforced learning process. Hence, we consider weighted combination as a better substitution of *mab*.

VII. CONCLUSION

KGA is an important task, which is critical to the management of knowledge. Existing methods rely on the labeled data to establish the connections among KGs. Nevertheless, these labeled entity pairs are actually hard to obtain in real-life scenarios. Hence, we put forward an active KGA framework,

KGA-SS, to address the scarce supervision issue. The two components in KGA-SS, i.e., RAS and CCL, can augment the supervision signals and lead to better alignment results. Extensive experiments on popular KGA benchmarks verify the effectiveness and generality of our proposed model.

As to future research directions, we plan to investigate from the following aspects. The first is KGA under the unsupervised setting. Although weakly supervised KGA can significantly reduce the requirement of labeled data, it still demands extra labeling effort given new data. Thus, it is important to explore